Akanksha Pandey, PhD

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SUMMARY

Computational biologist combining a PhD in phylogenomic methodology (structure-aware evolutionary modeling, clade-specific substitution models) with four years of industry experience integrating multi-omics data for oncology target discovery and drug-combination rationale. Research interests: evolutionary and statistical models for cancer genomics.

EDUCATION

Ph.D., Biology (Computational Biology) — University of Florida, Gainesville, FL
Advisor: Prof. Edward L. Braun, Department of Biology — 2013 – 2019

B.Tech., Bioinformatics — VIT University, Vellore, TN, India
2008 – 2012

RESEARCH EXPERIENCE

Bioinformatics Scientist • Kronos Bio, Inc. • Cambridge, MA — **01/2020 – 11/2023**

Developed bioinformatics pipelines for target identification, biomarker discovery, drug-combination rationale, and patient stratification in oncology drug discovery.

- Implemented a network-based multi-omics integration approach combining RNA-seq and ChIP-seq data to identify gene regulatory pathways underlying transcriptional deregulation, enabling identification of first-in-class, druggable targets.
- Built a machine-learning framework integrating high-throughput perturbation screening data with public omics resources to elucidate the mechanism of action of a CDK9 inhibitor, prioritizing ~15 genes for experimental validation.
- Designed and executed custom RNA-seq analyses characterizing transcriptional and pathway-level effects of a p300 inhibitor in HPV-driven cervical cancer, providing proof-of-concept evidence supporting therapeutic potential.
- Developed a scalable computational pipeline quantifying drug synergy from PRISM high-throughput combination screening data, identifying novel combination strategies and patient populations for an existing clinical asset (presented at ASH 2023).
- Led a large-scale external collaboration with the Broad Institute, coordinating analysis of >100 compounds across ~950 cancer cell lines and translating integrated datasets into a prioritized shortlist of ~20 compounds for downstream validation.
- Built and maintained reproducible NGS workflows using Python, R, Nextflow, Docker, and HPC/AWS environments supporting rigorous, high-throughput analysis across multiple research programs in parallel.
- Communicated complex analytical results through presentations, written reports, and cross-functional discussions, distilling key take-home messages and recommendations for next experimental steps.

Graduate Research Scientist • University of Florida • Gainesville, FL — **08/2013 – 05/2019**

Developed structure-aware evolutionary models and clade-specific substitution frameworks to resolve conflicting topologies in deep phylogenomics, with a focus on the metazoan root.

- Built a structure-aware phylogenomic pipeline combining per-site solvent accessibility and secondary-structure prediction (SCRATCH-1D) with site-heterogeneous substitution models; showed that exposed and buried residues recover different metazoan-root topologies and that amino-acid recoding reduces the conflict (Pandey & Braun, Biology 2020).
- Trained 14 new clade-specific protein substitution models across seven deep-metazoan datasets spanning birds, mammals, plants, oomycetes, and yeasts; linked shifts in amino-acid exchangeability to long-term effective population size and released open-source models and training pipeline (Pandey & Braun, ACM BCB 2020).
- Developed per-site likelihood diagnostics ("strongly decisive sites") to quantify how taxon sampling and model complexity shape support for competing metazoan-root topologies under LG, GTR20, and LGL profile-mixture models, benchmarked in RAxML, IQ-TREE, and PAUP* on CIPRES and local HPC (Pandey & Braun, Biophysica 2021).
- Contributed to the Avian Phylogenomics Consortium, developing pipelines to extract and align orthologous markers from low-quality whole-genome assemblies and analyzing how data type and taxon sampling shape the avian tree of life (Reddy et al., Systematic Biology 2017; Tiley, Pandey et al., Avian Research 2020).
- Analyzed genome-wide sequencing data from Florida's coastal seaside sparrows to quantify population structure and inform conservation management (Enloe, Cox, Pandey et al., Conservation Genetics 2022).

Independent Research **11/2023 – Present**

Independent computational projects in cancer genomics, proteomics, and single-cell analysis.

- Conducted a case-control GWAS of 3,364 candidate germline variants for early-onset pancreatic cancer using PLINK2, identifying a lead association in PSMA8 (OR=2.61, p=9.3e-12) from a CRISPR-nominated variant panel on chromosome 18.
- Performed exploratory proteomics analysis of 2,832 Olink Explore HT proteins from the Ornish Alzheimer's lifestyle intervention trial, identifying 33 differentially abundant proteins (FDR < 0.05) with pathway enrichment and longitudinal modeling.
- Benchmarked public 10x Genomics scRNA-seq datasets using scanpy (QC, integration, clustering, differential expression) to build fluency in modern single-cell workflows.
- Developed a modular, reproducible RNA-seq pipeline (QC → alignment → quantification → differential expression) using Nextflow, deployable across local, cloud, and HPC environments.
- Implemented agentic AI systems using LangChain, LangGraph, and CrewAI, including custom tools for LLM-powered agents with Pydantic validation and stateful orchestration.

PUBLICATIONS

- Enloe, C., Cox, W.A., Pandey, A., Taylor, S.S., Woltmann, S., Kimball, R.T. (2022). Genome-wide assessment of population structure in Florida's coastal seaside sparrows. *Conservation Genetics 23(2), 285–297.* https://doi.org/10.1007/s10592-021-01415-5
- Pandey, A., Braun, E.L. (2021). The roles of protein structure, taxon sampling, and model complexity in phylogenomics: a case study focused on early animal divergences. *Biophysica 1(2), 87–105.* https://doi.org/10.3390/biophysica1020008
- Pandey, A., Braun, E.L. (2020). Protein evolution is structure dependent and non-homogeneous across the tree of life. *Proceedings of the 11th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics (BCB '20), 28, 1–11.* https://doi.org/10.1145/3388440.3412473
- Pandey, A., Braun, E.L. (2020). Phylogenetic analyses of sites in different protein structural environments result in distinct placements of the metazoan root. *Biology 9(4), 64.* https://doi.org/10.3390/biology9040064

- Tiley, G.P., Pandey, A., Kimball, R.T., Braun, E.L., Burleigh, J.G. (2020). Whole genome phylogeny of Gallus: introgression and data-type effects. *Avian Research 11(1), 7.* https://doi.org/10.1186/s40657-020-00194-w
- Reddy, S., Kimball, R.T., Pandey, A., Hosner, P.A., Braun, M.J., Hackett, S.J., et al. (2017). Why do phylogenomic data sets yield conflicting trees? Data type influences the avian tree of life more than taxon sampling. *Systematic Biology 66(5), 857–879.* https://doi.org/10.1093/sysbio/syx041

CONFERENCE PRESENTATIONS & POSTERS

- Kobylarz, M.J., Larke, M.S.C., Glore, W.L., Shum, M.G., Li, J., Mori, K.R., Cohen, E.B., Pandey, A., et al. Oncogenic human papillomavirus hijacks p300 to drive viral transcription, creating a therapeutic vulnerability that can be exploited with selective p300/CBP catalytic inhibitors. AACR-NCI-EORTC International Conference on Molecular Targets and Cancer Therapeutics, 2024.
- Pandey, A., Carvajal, L.A., Boghossian, A.S., Rees, M.G., Ronan, M.M., Roth, J.A., et al. The novel Profiling Relative Inhibition Simultaneously in Mixtures (PRISM) platform identifies synergistic activity of lanraplenib and ruxolitinib in hematological malignancies. American Society of Hematology (ASH) Annual Meeting, 2023.
- Day, M.A.L., Saffran, D.C., Hood, T., Noe, C., Naffar-Abu Amara, S., Singhania, A., Pandey, A., et al. KB-0742 is active in preclinical MYC-high models of TNBC, ovarian, and DLBCL cancers. AACR Annual Meeting, 2022.
- Day, M.A.L., Saffran, D.C., Hood, T., Obholzer, N., Pandey, A., et al. CDK9 inhibition via KB-0742 is a potential strategy to treat transcriptionally addicted cancers. AACR Annual Meeting, 2022.
- Day, M.A.L., Saffran, D.C., Hood, T., Obholzer, N., Pandey, A., et al. CDK9 inhibitor KB-0742 is active in preclinical models of small-cell lung cancer. AACR Annual Meeting, 2022.
- Day, M.A.L., Obholzer, N.D., Pandey, A., et al. CDK9 inhibition by KB-0742 is selective for transcriptionally addicted tumors harboring MYC genomic amplifications. AACR Annual Meeting, 2021.

TECHNICAL SKILLS

- **Programming & Infrastructure:** Python (SciPy, Pandas, NumPy, Matplotlib, Seaborn), R (Bioconductor, Tidyverse), shell/bash, Nextflow, Docker, AWS, HPC, Git, Linux/Unix
- **Phylogenetics & Evolutionary Modeling:** RAxML, IQ-TREE, PAUP*, PhyloBayes, PMSF, LG/GTR20/CAT/LGL mixture models, SCRATCH-1D, TopCons, MAFFT, CIPRES
- **Bioinformatics & NGS:** SAMtools, STAR, Bowtie, MACS2, DESeq2, EdgeR, scanpy, CellRanger, Salmon, PLINK2, Olink Explore HT; workflows for RNA-seq, ChIP-seq, scRNA-seq, GWAS, and multi-omic integration
- **Machine Learning:** scikit-learn, PyTorch, dimensionality reduction (PCA, UMAP), hypothesis testing, feature engineering for high-throughput screening data
- **LLMs & Agentic AI:** LangChain, LangGraph, CrewAI, Pydantic validation, GitHub Copilot, Claude Code with VS Code
- **Public Resources & Datasets:** TCGA, CCLE, DepMap, COSMIC, cBioPortal, GTEx, GDSC, Tempus AI

SELECTED CERTIFICATIONS

- Machine Learning, Stanford University (Coursera)
- Building AI Agents and Agentic Workflows Specialization (Coursera)
- Functions, Tools and Agents with LangChain, DeepLearning.AI
- Fundamentals of AI Agents Using RAG and LangChain (Coursera)